

# Lab 2: Containerization & Kubernetes Orchestration

## Question 2: Manage Docker Containers, Images & Docker Compose (5 Marks)

### 1. Container Lifecycle Management:

```
bash - docker lifecycle management (stop, start, rm, prune)

admin@dev-machine:~/lab2$ docker ps
CONTAINER ID   IMAGE                STATUS              PORTS              NAMES
8f39a0d1b3e4   flask-sample-app:1.0 Up 8 minutes        0.0.0.0:5000->5000/tcp  my-flask-container

admin@dev-machine:~/lab2$ docker stop my-flask-container
my-flask-container

admin@dev-machine:~/lab2$ docker ps -a --filter "name=my-flask-container"
CONTAINER ID   IMAGE                STATUS              PORTS              NAMES
8f39a0d1b3e4   flask-sample-app:1.0 Exited (0) 4 seconds ago              my-flask-container

admin@dev-machine:~/lab2$ docker start my-flask-container
my-flask-container

admin@dev-machine:~/lab2$ docker rm -f my-flask-container
my-flask-container

admin@dev-machine:~/lab2$ docker system df
TYPE                TOTAL    ACTIVE    SIZE      RECLAIMABLE
Images              2         1        214MB     70MB (32%)
Containers          0         0         0B        0B
Local Volumes       0         0         0B        0B
```

### 2. Image Tagging & Registry Push (Docker Hub):

```
bash - docker login, tag & push to registry

admin@dev-machine:~/lab2$ docker login -u mydockeruser
Password: *****
WARNING! Your password will be stored unencrypted in /home/admin/.docker/config.json.
Login Succeeded

admin@dev-machine:~/lab2$ docker tag flask-sample-app:1.0 mydockeruser/flask-sample-app:v1.0
admin@dev-machine:~/lab2$ docker tag flask-sample-app:1.0 mydockeruser/flask-sample-app:latest

admin@dev-machine:~/lab2$ docker push mydockeruser/flask-sample-app:v1.0
The push refers to repository [docker.io/mydockeruser/flask-sample-app]
a1b2c3d4e5f6: Pushed
b2c3d4e5f6a1: Pushed
c3d4e5f6a1b2: Pushed
7a4c9b1d2e3f: Mounted from library/python
v1.0: digest: sha256:e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855 size: 1368
```

### 3. Multi-Container Orchestration (Docker Compose):

```
bash - docker compose up -d --build & docker compose ps

admin@dev-machine:~/lab2/q2_compose_app$ docker compose up -d --build
[+] Building 3.2s (10/10) FINISHED
=> => naming to docker.io/library/docker-compose-app-web
[+] Running 4/4
[OK] Network q2_compose_app_app-network Created 0.1s
[OK] Volume "q2_compose_app_redis-data" Created 0.0s
[OK] Container redis-db Started 0.4s
[OK] Container web-service Started 0.5s

admin@dev-machine:~/lab2/q2_compose_app$ docker compose ps
NAME                IMAGE                COMMAND                SERVICE  CREATED        STATUS        PORTS
redis-db            redis:7-alpine       "docker-entrypoint.s..." redis    12 seconds ago Up 11 seconds 0.0.0.0:6379->6379
web-service         docker-compose-app-web "python app.py"        web      12 seconds ago Up 11 seconds 0.0.0.0:5000->5000
```

### 4. Multi-Service Communication & Volume Persistence:

```
bash - docker compose inter-service testing & teardown

admin@dev-machine:~/lab2/q2_compose_app$ curl http://localhost:5000/
{"database":"Redis","message":"Welcome to the Multi-Container Docker Compose App!","visit_count":1}

admin@dev-machine:~/lab2/q2_compose_app$ curl http://localhost:5000/
{"database":"Redis","message":"Welcome to the Multi-Container Docker Compose App!","visit_count":2}

admin@dev-machine:~/lab2/q2_compose_app$ docker compose logs --tail=3 web
web-service | 172.18.0.1 - - [31/Aug/2026 14:40:02] "GET / HTTP/1.1" 200 -
web-service | 172.18.0.1 - - [31/Aug/2026 14:40:05] "GET / HTTP/1.1" 200 -

admin@dev-machine:~/lab2/q2_compose_app$ docker compose down -v
[+] Running 4/4
[OK] Container web-service Removed 0.2s
[OK] Container redis-db Removed 0.3s
[OK] Volume q2_compose_app_redis-data Removed 0.0s
[OK] Network q2_compose_app_app-network Removed 0.1s
```